EXP NO:- 4

PROGRAM ➖

```
clc;
clear;
close all;
N = 4;
Fs = 1000;
Fc = 100;
Wn = Fc/(Fs/2);
Rp = 1;
Rs = 40;
% Butterworth Filter
[b1,a1] = butter(N,Wn,'low');
% Chebyshev Type-I Filter
[b2,a2] = cheby1(N,Rp,Wn,'low');
% Elliptic Filter
[b3,a3] = ellip(N,Rp,Rs,Wn,'low');
% Frequency Response Comparison
[H1,f] = freqz(b1,a1,512,Fs);
[H2,~] = freqz(b2,a2,512,Fs);
[H3,~] = freqz(b3,a3,512,Fs);
plot(f,20*log10(abs(H1)+eps),'b');
hold on;
plot(f,20*log10(abs(H2)+eps),'r');
plot(f,20*log10(abs(H3)+eps),'g');
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Comparison of IIR Filters');
legend('Butterworth','Chebyshev-I','Elliptic');
ylim([-100 5]);
```

```
disp('Butterworth Coefficients:');
disp(b1); disp(a1);
disp('Chebyshev-I Coefficients:');
disp(b2); disp(a2);
disp('Elliptic Coefficients:');
disp(b3); disp(a3);
```

OUTPUT:-

